

Scene Configuration Reference

Use this as the exact wiring reference for `GameScene.unity`.

Root Hierarchy (recommended)

- Main Camera
- GameSystems
- GameManager (component)
- SpawnManager (component)
- AudioManager (component)
- Environment
- ParallaxRoot (ParallaxBackground)
 - BgLayerFar (SpriteRenderer)
 - BgLayerNear (SpriteRenderer)
- StarFieldRoot (StarField)
- PlayerShip (PlayerController)
- FirePoint
- ShieldVisual (optional, disabled by default)
- Canvas
- MainMenuPanel
- HUDPanel
- PausePanel
- GameOverPanel
- WaveBannerPanel
- UIRoot (UIManager)
- EventSystem

Component Reference Wiring

SpawnManager

- Basic Enemy Prefab -> EnemyBasic prefab
- Zigzag Enemy Prefab -> EnemyZigzag prefab
- Tank Enemy Prefab -> EnemyTank prefab

PlayerController

- Bullet Prefab -> PlayerBullet prefab
- Fire Point -> PlayerShip/FirePoint
- Shield Visual -> PlayerShip/ShieldVisual (optional)
- Explosion Prefab -> Explosion prefab (optional)

EnemyController (per enemy prefab)

- Bullet Prefab -> EnemyBullet prefab
- Fire Point -> prefab child `FirePoint`

- PowerUp Prefabs -> [PowerUpHealth, PowerUpWeapon, PowerUpShield]
- Explosion Prefab -> Explosion prefab (optional)

UIManager

Assign all panel and UI references from Canvas children:

- Main menu fields/buttons
- HUD texts + slider + icons
- Pause menu buttons
- Game over texts/buttons
- Wave banner panel and text

Scene Defaults

- Game starts in Main Menu state.
- **Start** button transitions to active play and first wave.
- **Escape** pauses/resumes.
- Player death transitions to Game Over.
- **Restart** resets gameplay state and starts from wave 1.